STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 3: Motion in a Plane

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Vector.

Suggested allocation by section: 1-mark questions earn one mark for the correct fact, formula or unit; 2-mark questions normally award one mark for the principle/formula and one for the explanation or result; 3-mark questions award credit for the relation, working/reasoning and result with unit; 4-mark questions award credit for the principle, equation, working and final answer/interpretation. Accept equivalent correct methods.

Q2 1 mark

Acostheta.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q3 1 mark

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q4 1 mark

Inward.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q5 1 mark

5.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q6 1 mark

v=romega.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q7 1 mark

45 deg .

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q8 1 mark

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q9 2 marks

A=5 N; A=8.66 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q10 2 marks

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q11 2 marks

24 m/s^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q12 2 marks

2 s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q13 2 marks

10.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q14 2 marks

4pi rad/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q15 3 marks

T=2 s; H=5 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q16 3 marks

10 N at 36.9 deg north of east.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q17 3 marks

4 m/s^2 inward.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q18 3 marks

3 s; 60 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q19 4 marks

2 s; 5 m; 34.6 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q20 4 marks

40 N inward; centripetal force does zero work.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.